Chapter 3

Big Data Architecture

Learning Objectives

- Appreciate the variety of architectures for Big Data ecosystem
- Explain the many layers of technologies in the Big Data architecture
- Analyse how leading organizations are architecting Big Data solutions

Introduction

Big Data Application Architecture is the configuration of tools and technologies to accomplish the whole task. An ideal Big Data architecture would be resilient, secure, cost-effective, and adaptive to new needs and environments. This can be achieved by beginning with a proven architecture, and creatively and progressively restructuring it as additional needs and problems arise. Big Data architectures should ultimately align with the architecture of the Universe itself, the source of all invincibility.

Caselet

Google Query Architecture

Google invented the first Big Data architecture. Their goal was to gather all the information on the web, organize it, and search it for specific queries from millions of users. An additional goal was to find a way to monetize this service by serving relevant and prioritized online advertisements on behalf of clients.

Google developed web crawling agents which would follow all the links in the web and make a copy of all the content on all the webpages it visited. Google invented cost-effective, resilient, and fast ways to store and process all that exponentially growing data. It developed a scale-out architecture in which it could linearly increase its storage capacity by inserting additional computers into its computing network. The data files were distributed over the large number of machines in the cluster. This distributed files system was called the Google File system, and was the precursor to Hadoop.

Google would sort or index the data thus gathered so it can be searched efficiently (Figure 3.1). They invented the key-pair NoSQL database architecture to store variety of data objects. They developed the storage system to avoid updates in the same place. Thus the data was written once, and read multiple times.

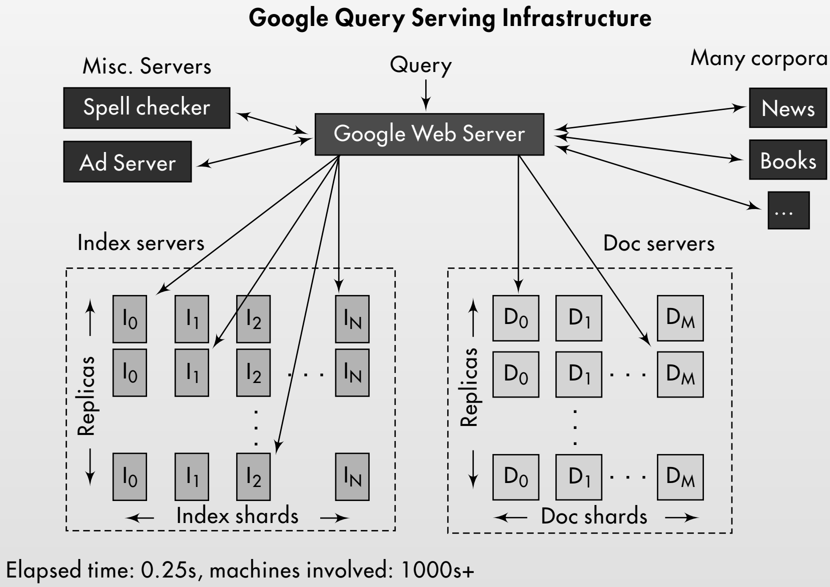


FIGURE 3.1 Google query architecture

Google developed the MapReduce parallel processing architecture whereby large datasets could be processed by thousands of computers in parallel, with each computer processing a chunk of data, to produce quick results for the overall job. The Hadoop ecosystem of data management tools like Hadoop distributed file system (HDFS), columnar database system like HBase, a querying tool such as Hive, and more, emerged from Google's inventions. Apache Spark is a streaming data technology to produce instant results.

1. Why should Google publish its File System and the MapReduce parallel programming system and send it into open-source system?
2. What else good can be done with Google's repository of the entire web's data?

3.1 STANDARD BIG DATA ARCHITECTURE

Here is the generic Big Data Architecture as introduced in Chapter 1 (shown again in Figure 3.2). There are many sources of data. All data is funneled in through an ingest system. The data is forked into two sides: a stream processing system and a batch processing system. The outcome of these processing can be sent into NoSQL databases for later retrieval, or sent directly for consumption by many applications and devices.

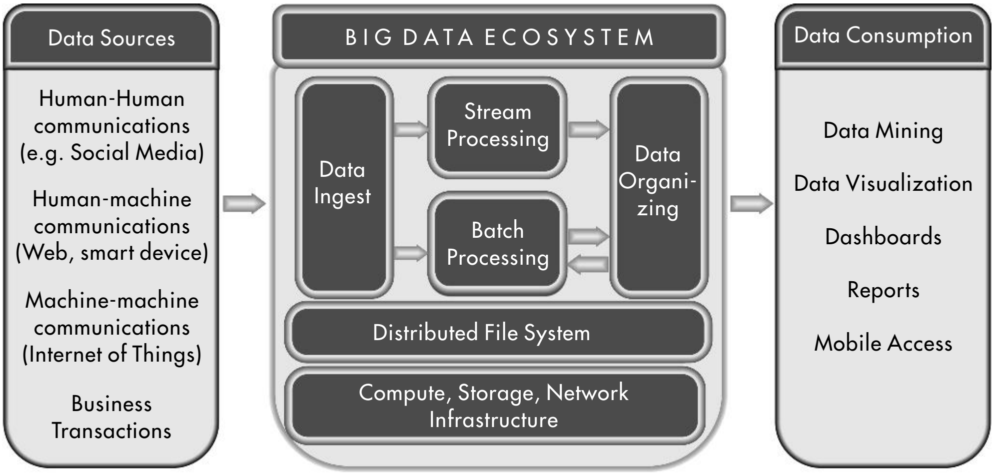


FIGURE 3.2 Generic big data architecture

A Big Data solution typically comprises these as logical layers. Each layer can be represented by one or more available technologies.

Big Data sources layer The choice of sources of data for an application depends upon what data is required to perform the kind of analyses you need. Big Data will vary in origin, size, speed, form, and function, as described by the 4 Vs in Chapter 1. The sources of Big Data, as described in Chapter 2, can be internal or external to the organization. The scope of access to data available could be limited. The level of structure as well as the speed of data and its quantity could be high or low depending upon the data source.

Data Ingest layer This layer is responsible for acquiring data from the various data sources. Incoming data is received through a scalable number of input points, that can acquire data at various speeds and in various quantities. The data is sent to a batch processing system, or a realtime processing system, or to a storage file system such as Hadoop. Compliance regulations and governance policies impact what data can be stored and for how long.

Batch Processing layer The analysis layer receives data from the ingest point or from the file system or from the NoSQL databases. Data is processed using parallel programming techniques (such as MapReduce) to process it and produce the desired results. This batch processing layer thus must understand the data sources and data types, the algorithms that would work on that data, and the format of the desired outcomes. The output of this layer could be sent for instant reporting, or be stored in NoSQL databases for an on-demand report, for the client.

Stream Processing layer This technology layer receives data directly from the ingest point. Data is processed using parallel programming techniques (such as MapReduce) to process it in real time, and produce the desired results. This layer thus should understand the data sources and data types extremely well, and the super-light algorithms that would work on that data to produce the desired results. The outcome of this layer too could be stored in the NoSQL Databases.

Data Organizing Layer This layer receives data from both the batch and stream processing layers. Its objective is to organize the data for easy access. It is represented by NoSQL databases. There are a variety of NoSQL databases to suit different needs. SQL-like languages like Hive and Pig can be used to access data easily and generate reports from these databases.

Infrastructure Layer At bottom there is a layer that manages the raw resources of storage, computation, and communication. This is increasingly provided through a cloud computing paradigm.

Distributed File System Layer This is the heart of a Big Data system. It would store huge quantities of data and make it quickly and securely available and accessible to the other layers. Hadoop Distributed File System (HDFS) is the primary technology in this layer. It would also include supporting applications, such as YARN (Yet Another Resource Manager) that enable the efficient access to data storage and its transfer.

Data Consumption layer This is the final layer, and it consumes the output provided by the analysis layers, directly or through the organizing layer. The outcome could be standard reports, data analytics, dashboards and other visualization applications, recommendation engine, on mobile and other devices.

3.2 BIG DATA ARCHITECTURE EXAMPLES

Every major organization and its applications have a unique optimized infrastructure to suit its specific needs. Some architecture examples from some very prominent users and designers of Big Data applications are as follows.

3.2.1 IBM Watson

IBM Watson uses Spark to manage incoming data streams. It also uses Spark's Machine Learning library (MLLib) to analyze data and predict diseases (Figure 3.3).

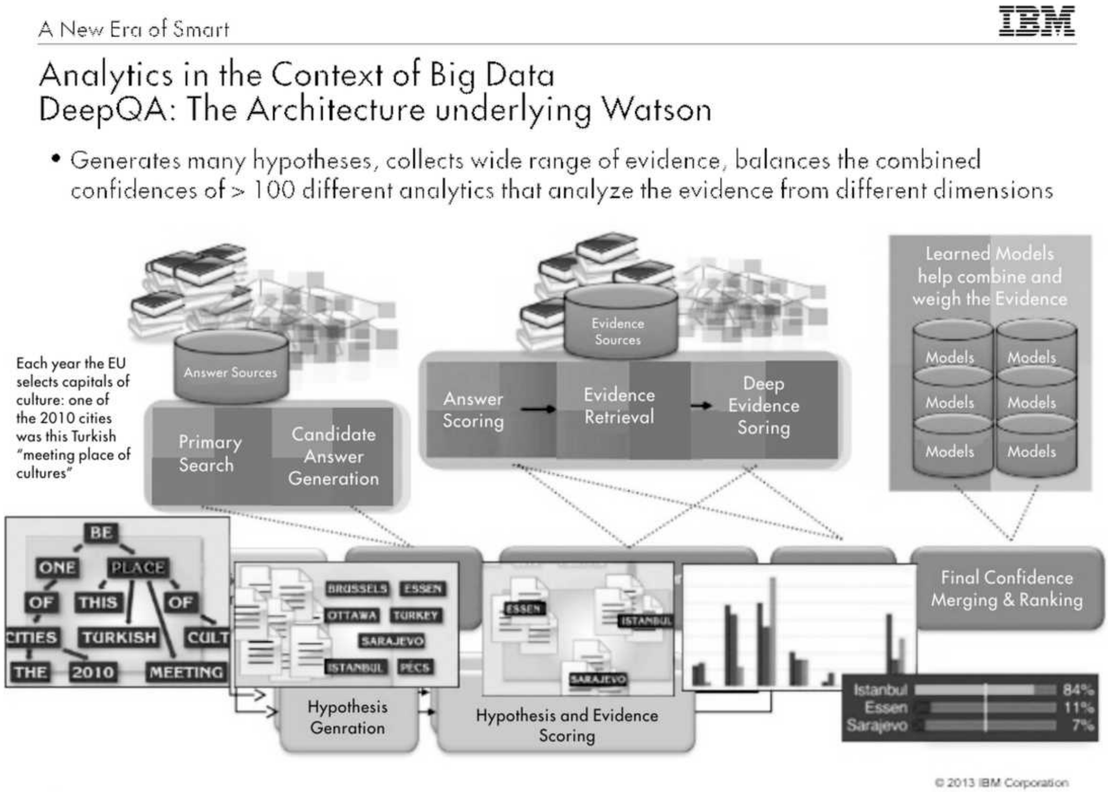


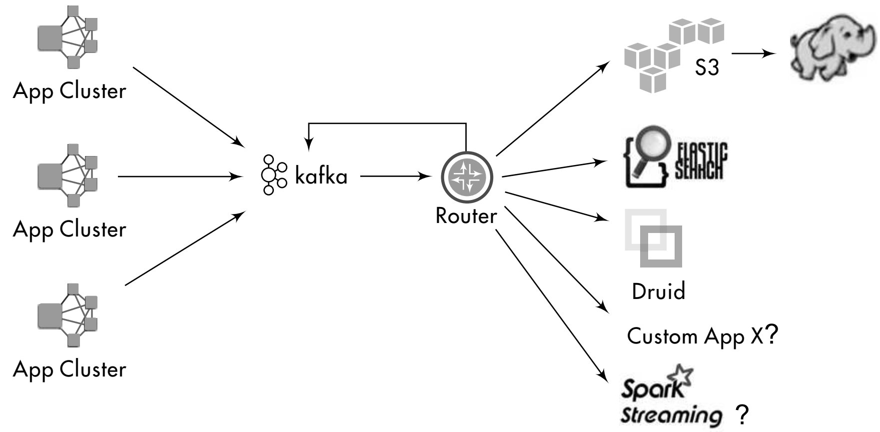
FIGURE 3.3 The architecture underlying IBM Watson

3.2.2 Netflix

This is one of the largest providers of online video entertainment. They handle more than 400 billion online events per day. As a cutting-edge user of Big Data technologies, they are constantly innovating their mix of technologies to deliver the best performance. They use Apache Kafka as the common messaging system for all incoming requests. They use Spark for stream processing and host their entire infrastructure on Amazon Web Services (AWS) cloud service (Figure 3.4). The database system is AWS' S3 as well as Cassandra and Hbase for storing data.

3.2.3 eBay

eBay is the second-largest e-commerce company in the world. It delivers 800 million listings from 25 million sellers to 160 million buyers. To manage this huge stream of activity, eBay uses a stack of Hadoop, Spark, Kafka, and other elements (Figure 3.5). They think that Kafka is the best new thing for processing data streams.



(*Source:* Netflix)

FIGURE 3.4 Netflix database processing and storage system

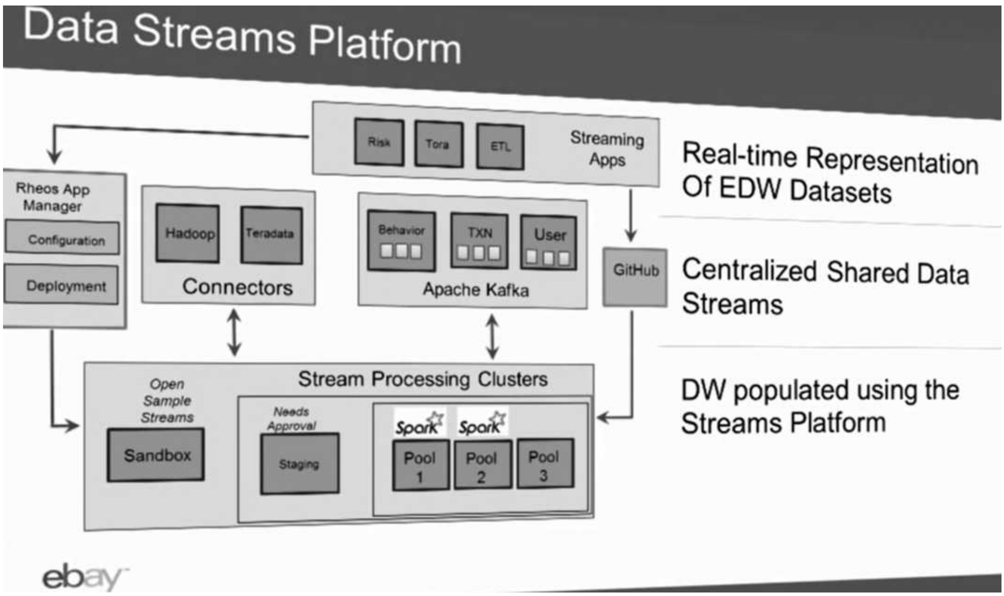


FIGURE 3.5 eBay data stream platforms

3.2.4 VMWare

VMware is a leading provider of virtualization systems used in cloud computing systems. VMware's view of a Big Data architecture is similar to, but more detailed, than our main big architecture diagram (Figure 3.6).

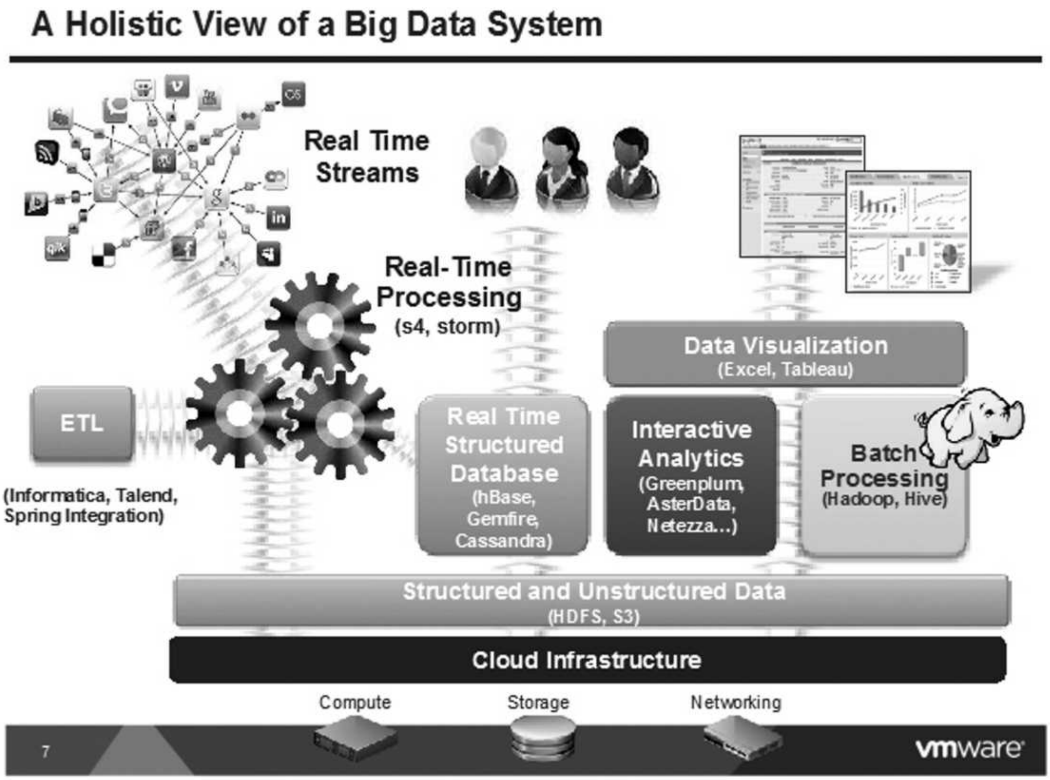


FIGURE 3.6 A holistic view of a big data system in VMWare

3.2.5 The Weather Company

The Weather company serves tens of billions of weather data requests globally through websites and mobile apps, everyday. It uses streaming architecture using Apache Spark to frame its Big Data infrastructure (Figure 3.7).

3.2.6 TicketMaster

This is the world's largest company that sells event tickets. Their goal is to make tickets available to real fans for purchase, and prevent bad actors from manipulating the system to increase the price of the tickets in the secondary markets. They use a mix of Hadoop, Spark, and Kafka systems to manage throughput and reliability (Figure 3.8).

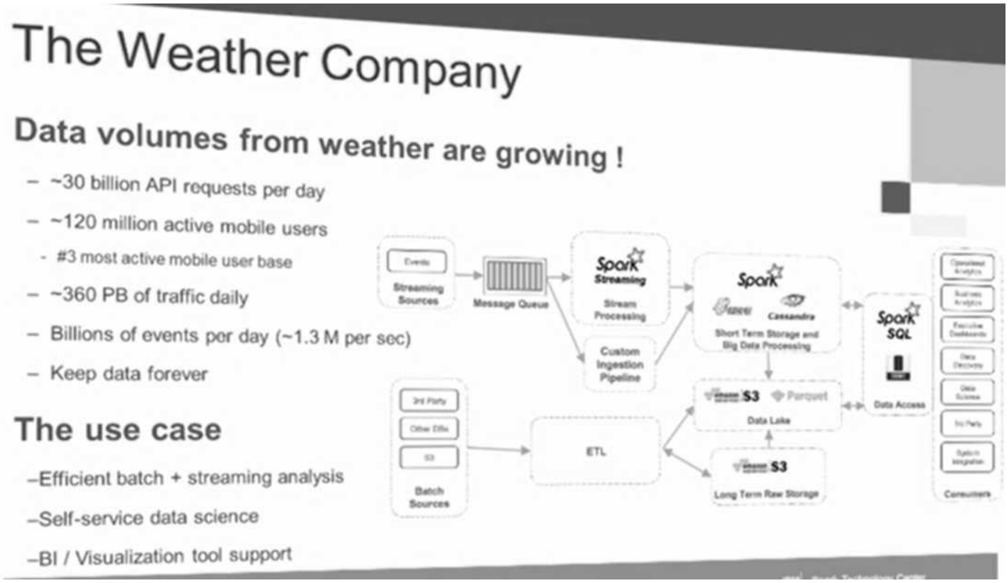


FIGURE 3.7 Streaming architecture using Apache Spark to frame Big Data infrastructure of the weather company

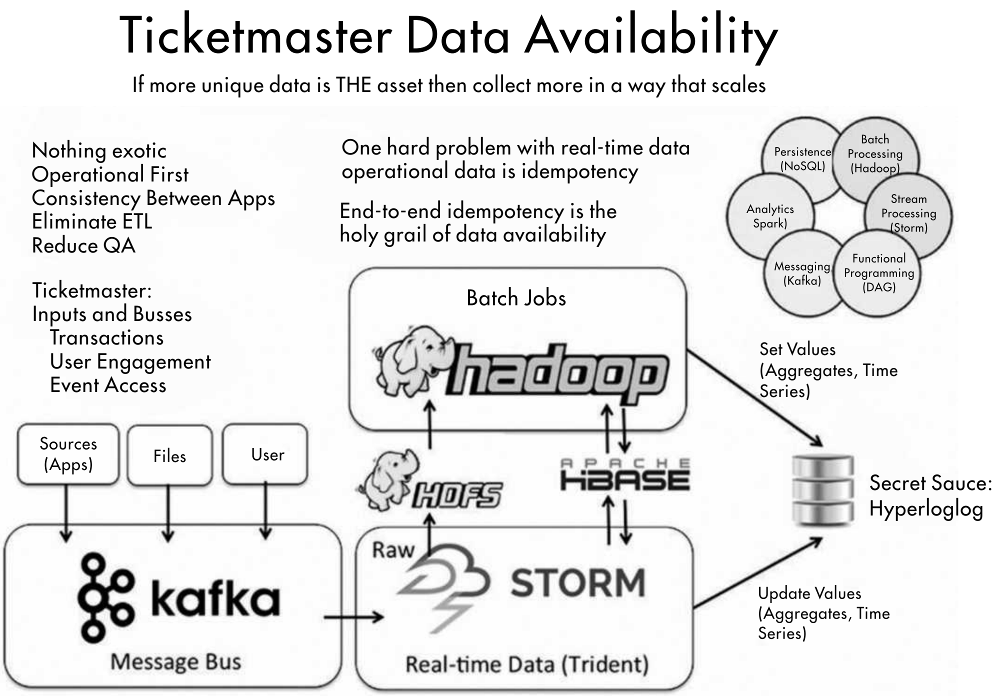


FIGURE 3.8 Mix of Hadoop, Spark, and Kafka systems to manage throughput and reliability by TicketMaster

3.2.7 LinkedIn

This professional networking company aims to maintain an efficient system for processing streaming data and make the link options available in real-time. They use Hadoop and Kafka for receiving and storing data, and Samza for stream processing (Figure 3.9).

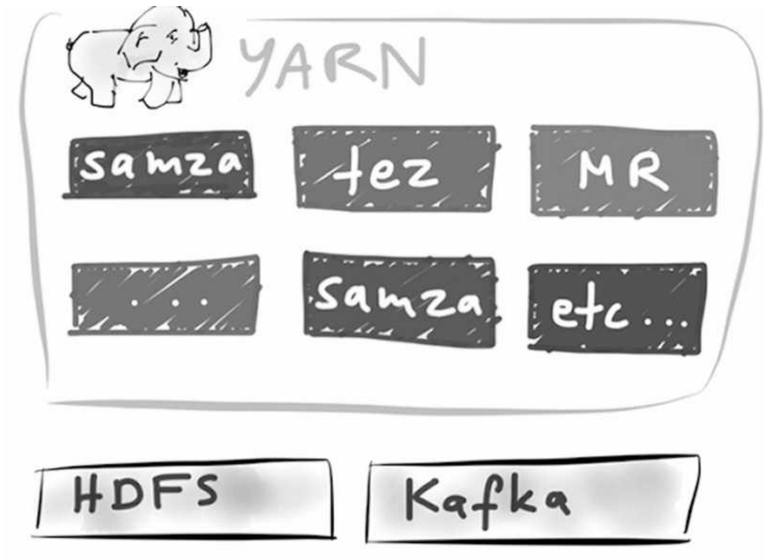


FIGURE 3.9 Mix of Hadoop, Spark, and Kafka systems to manage data storage and Samza for stream processing by LinkedIn

3.2.8 Paypal

This payments-facilitation company needs to understand and acquire customers, and process a large number of payment transactions. They use Apache Storm for stream processing of data. They use Druid for distributed data storage. They also use Apache Titan Graphical (NoSQL) database to manage links and relationship among data elements (Figure 3.10).

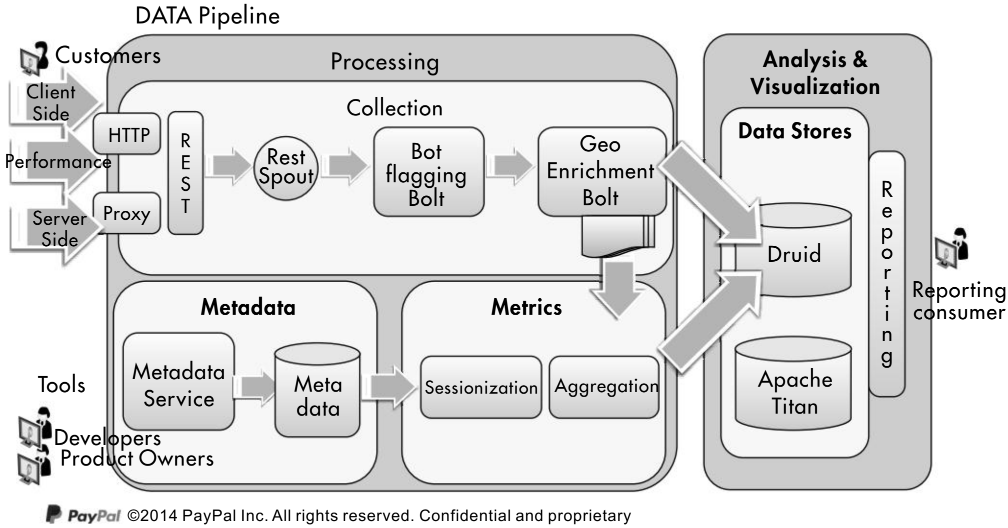


FIGURE 3.10 Apache Storm and Apache Titan Graphical (NoSQL) database to manage links and relationship among data elements in Paypal

3.2.9 CERN

This premier high-energy physics research lab computes petabytes of data using in-memory stream processing to process data from millions of sensors and devices (Figure 3.11).

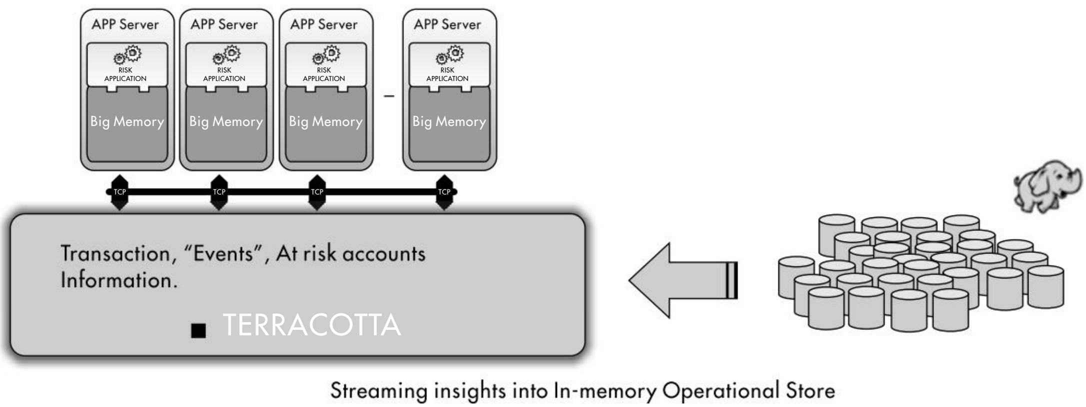


FIGURE 3.11 Streaming insights into In-memory operational store in CERN lab

3.3 CONCLUSION

Big Data architectures are multi-tier designs that use technologies for ingesting, storing, processing, and delivering data to user applications. Data is ingested and fed into both streaming and batch processing engines. Most tools used for big data processing are open source tools served through the Apache community.

Review Questions

1. Describe the Big Data processing architecture.
2. What are Google's contributions to Big data processing?
3. What are some of the hottest technologies visible in Big Data processing?

True/False Questions

1. The more the sources of Big data included, the more complex will the Big Data architecture be.
2. Big Data architecture includes many kinds of data sources.
3. Big Data architecture includes transaction processing applications.
4. A simple Big Data architecture consists of two layers: a data ingest layer, and a data consumption layer.
5. Once established, a company's Big Data Architecture cannot be changed.
6. Serving web-based apps is a form of a Big Data application.
7. Google invented Big Data architecture with Search Query engine.
8. IBM Watson uses Big Data architecture to do Deep Question-Answer applications.
9. Facebook is a not an active user of Big Data architectures.
10. Government organizations should use a more secure Big Data architecture.

Liberty Stores Case Exercise: Step B3

Liberty wants to build a scalable and futuristic listening platform for understanding its customers and other stakeholders.

1. What kind of Big Data architecture would you suggest for this company?